



A.K.D.DHARMA RAJA WOMEN'S COLLEGE
(Affiliated to Madurai Kamaraj University)
RAJAPALAIYAM

DEPARTMENT OF MATHEMATICS (B.Sc)

Course Outcomes (COs)

CO 1- Apply Mathematical Concepts

Understand and apply fundamental concepts of calculus, algebra, differential equations, and linear algebra to solve real-world and theoretical problems.

CO 2- Develop Analytical Thinking

Enhance logical reasoning and analytical skills through the study of abstract mathematics and problem-solving techniques.

CO 3 - Use Mathematical Tools and Techniques

Gain proficiency in using mathematical software and tools (like MATLAB, Mathematica, or R) for modeling, simulations, and complex computations.

CO 4 -Formulate and Solve Problems

Translate physical, social, and economic problems into mathematical models and analyze them for viable solutions.

CO 5- Communicate Mathematical Ideas

Present mathematical arguments and conclusions clearly and effectively, both in writing and orally.

Programme Outcomes (POs)

PO1: Disciplinary Knowledge

Demonstrate comprehensive knowledge and understanding of core mathematical concepts, including calculus, algebra, geometry, and statistics.

PO2: Problem Solving and Analytical Thinking

Analyze and solve complex mathematical problems using logical reasoning, critical thinking, and appropriate mathematical tools.

PO3: Computational Skills

Apply appropriate computational techniques and use mathematical software to model, analyze, and interpret data and equations.

PO4: Communication Skills

Communicate mathematical concepts and solutions clearly and effectively in written, oral, and symbolic forms.

PO5: Research and Inquiry

Develop the ability to engage in independent research, apply mathematical methods to real-life scenarios, and interpret results logically.

PO6: Modern Tool Usage

Utilize modern tools, software, and technology for solving mathematical problems and for conducting simulations or experiments.

PO7: Ethics and Professional Responsibility

Apply ethical principles and commit to professional responsibilities in academic and applied mathematical contexts.

PO8: Lifelong Learning

Develop the ability to pursue continuous learning and adapt to the evolving trends in mathematics and related disciplines.

PO9: Interdisciplinary Applications

Apply mathematical knowledge and techniques to solve problems in allied fields such as physics, computer science, economics, and engineering.

PO10: Teamwork and Leadership

Demonstrate the ability to work effectively in teams, take initiative, and develop leadership qualities in academic, research, and professional environments.

Programme Specific Outcomes (PSOs)**PSO1: Mastery of Core Mathematical Concepts**

Acquire in-depth knowledge of core areas of mathematics such as algebra, calculus, real and complex analysis, number theory, differential equations, and linear algebra.

PSO2: Application of Mathematics in Other Disciplines

Apply mathematical techniques and reasoning in interdisciplinary contexts such as physics, computer science, economics, and engineering.

PSO3: Mathematical Modeling and Problem Solving

Formulate real-world problems as mathematical models and use analytical and numerical methods to find effective solutions.

Programme Educational Objectives (PEOs)**PEO1: Foundation in Mathematical Sciences**

Provide students with a solid foundation in pure and applied mathematics to understand and solve complex mathematical problems.

PEO2: Higher Education and Research

Prepare graduates for successful admission into postgraduate programs, research, and professional courses in mathematics and related disciplines.

PEO3: Employability and Professional Development

Equip students with analytical, logical, and quantitative skills to pursue careers in education, data analysis, actuarial science, finance, and other mathematics-intensive professions.

